USE BLACK BALL POINT PEN ONLY

UG/MBA/MCA

ವಿಶ್ವೇಶ್ವರಯ್ಯ ತಾಂತ್ರಿಕ ವಿಶ್ವವಿದ್ಯಾಲಯ, ಬೆಳಗಾವಿ

VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI

Session	Morning ○	Afternoon ●
Subject	Computer networks	
Date	29/05/26	Semester V th

Script No.

U252851954

Subject Code: BCS502

University Seat Number: 1OB23C1034

I abide by the rules and regulations of the university
I have written USN and signed correctly

JAYANTH S

Name and Signature of the Candidate

I have verified the data filledby the Candidate of thisbooklet & found correct

29/05/26

Room Superintendents's Signature

of Graph Sheets / Drawing Sheets Enclosed

INSTRUCTIONS TO CANDIDATE

1. This form will be scanned by Computer. This Answer Book must contain 44 Pages.
2. Fill / Darken this form with BLACK INK BALL POINT PEN only.
3. Please darken the appropriate circles for USN, Date, Session and Questions answered.
4. Wrong darkening or double circle darkening will be treated as invalid and will lead to delay in processing of results.
5. The Candidate should follow instructions carefully. The candidate is responsible for any lapses in following the instructions.

INSTRUCTIONS TO THE ROOM SUPERINTENDENT

1. The Room Superintendent shall check that the USN and Signature of the candidate are filled correctly in appropriate boxes of the answer booklet before signing on the facing sheet.
2. If any discrepancy is found in the answer script like loose threading, less number of pages,wrong darkening of bubbles of USN/ Subject code should be entered in Form B against USN of candidate.
3. Corrections to mistakes in bubbling of USN or Subject code can be made using Whitener.

To be signed by the Chief Superintendent if student is booked under MPC	

INSTRUCTIONS TO CANDIDATES

1. **Only a single answer book will be issued. No additional Answer Books are permitted.**
2. **The candidate should write his / her seat number and give other information like examination, semester subject, subject code etc., against the space provided on the title page of the answer book.**
3. Revealing your identity in any other place of the answer booklet will be treated as Malpractice.
4. The question numbers should be written in the margin provided for the purpose.
5. The candidate shall **write answers on both the sides of pages** of the answer book. All **rough work** must be done in the **space provided** at the **end of the answer book.** Answers must be written using **black ball point pen only**. If there is **a change in pen,** the same shall be **attested** by the **Room Superintendent on the form B.**
6. **Answer book should be handed over personally to Room Superintendent before leaving the examination hall.**
7. No candidate shall be permitted to go to toilet during the period of examination.
8. The candidate should not take any books / notes, log table, scribbling pads, Cell phones, programmable calculators or any kind of reference material into the examination hall. The candidate should make sure that he / she has no unauthorized book or paper in the examination hall with him / her or in his / her desk. He / She should have only articles permitted like Identity Card, Hall Ticket / Admission Ticket. The Candidate should not write anything on the Admission Ticket or Identity Card or Calculator.
9. **All the Candidates should take possession of their seats 30 minutes before the commencement of the examination.** A warning bell will be given 10 minutes before the commencement of the examination. another bell be given at the beginning of the examination when question papers will be distributed and the candidate should start writing the answers. **No candidate shall be admitted after 30 minutes of the commencement of the examination and shall be allowed to leave the examination hall before 45 minutes of the commencement of the examination. No candidate should leave his / her seat during last 10 minutes.** Warning bell be given 10 minutes before the closing time and final bell is given at the end of the examination. Then all the candidates should stop writing or revising the answers and should handover the answer book to the Room Superintendent.
10. **The candidates should see that, the Room Superintendent has appended his / her signature at the specified space on the answer book, before he / she hands over the Answer Book to the Room Superintendent.**
11. Smoking and taking tea or coffee or cold drink in the examination hall is strictly prohibited. However, drinking water will be supplied on request.
12. **The Candidate should check the Answer Script for any discrepancy in threading, incorrect number of pages, missing of barcode etc. and should bring the same to the notice of Room Supervisor.**
13. **Any Candidate appearing for the UG / PG examination is liable to be charged with committing malpractice in the following cases :**
    a) **Bringing** in the examination hall or being found in possession of portions of a book, manuscript, Programmable Calculator or any other **material or matter,** which is **not permissible to be brought in to the examination hall.**
    b) **Having any written matter on scribbling pad, Question Paper, Admission Ticket, Calculator, any part of the body. Kerchief, Clothes, Socks, Instrument Box, Identity Card, Scales etc.,**
    c) **Copying** from the material or matter or answer of another candidate or similar aid or assistance is rendered to another candidate within the Examination Hall.
    d) **Communicating** with any candidate or any other person inside or outside the examination hall with a view to take assistance or aid to write answers in the examination.
    e) **Making any request of representation or offers any threat for inducement or inducing to bribery to Room Superintendent. or and any other official or officer of the University / College for favours** in the examination hall or to the Examiner in the answer script.
    f) **Smugglig out or smuggling in or tearing off of the answer script sheets or supplementary sheets or inserting** papers written outside the examination hall into the answer book or **running away along with answer script from the examination hall or premises.**
    g) **Impersonating** or allowing any other person to impersonate to answer in his / her place in the examination hall.
    h) **Supply of copying material** inside or from outside the examination hall.
    i) **Bringing mobile phone** to the examination hall.
    j) **Unruly behavior** inside or near the examination hall.

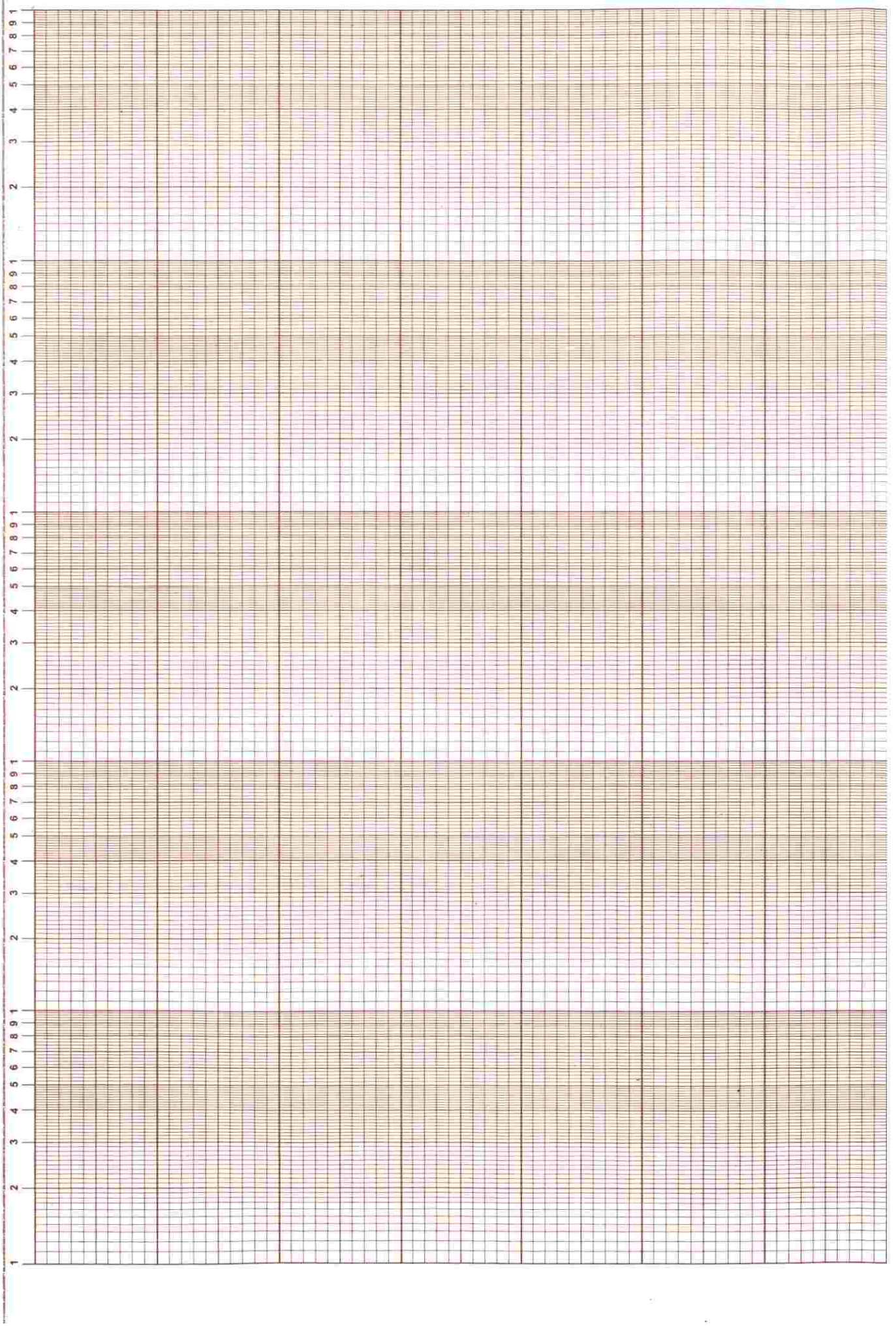
REMEMBER : YOUR FUTURE IS BRIGHT. DON'T SPOIL IT.

PUNISHABLE

Nos.

Q. Nos.

SEMI-LOG PAPER (5 CYCLES X 1/10")



05

Q. Nos.

Subject: Computer network.

Subject Code: BCS502

Questions Answered
(darken the circles for answered questions)

1	●	●	c	d	e	6	●	●	c	d	e
2	a	●	c	d	e	7	a	b	c	d	e
3	a	●	c	d	e	8	●	●	c	d	e
4	a	b	c	d	e	9	a	b	c	d	e
5	a	b	c	d	e	10	a	b	c	d	e

START ANSWERS FROM HERE

Q.1

a Answer:-

Data Communication:- Is a process in which the transmission of the data is being done from one point to another point within a network. where are five essential components through which the data is being transfred. these are mentioned below:-

message:- In this component of Data communication the Data is present which have to be transfred. the Data might exsist in any form, text, matrix images., Binary bits etc.

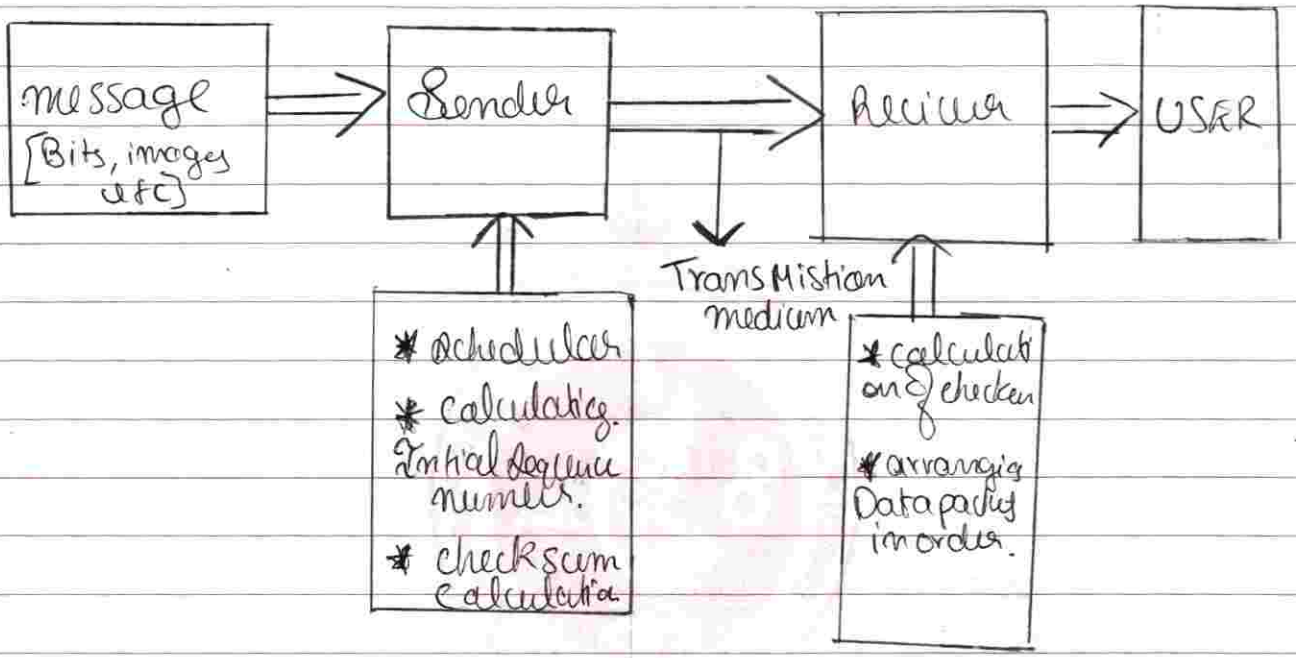
Sender:- This is the Component which is responsiable for initiating the Data transfer across the network.

reciver:- This is the Component which recieving the Data and verifies its corrupted or not and accepts.

transmition medium:- without this Component the Data transfer deesnot exsist where this is one of the main Component which provides a pathway to transfer a data from one point to another. Example:- guided & unguided medium.

Q. Nos.

* Protocol: This is the set of the rules that must have to be followed inorder to transmit the data without following the protocal the transfered data is just an ordinary electronic signals which is of no use.



The three criteria's necessary for an effective & efficient network.,

* Error detation:- due to the noise in the trans-misstion medium the Data packets get corrupted and cause the error or Damages the Data to avoid this acceptance & the corrupted data, establish the virtual connection circuit where the Data packets dosnot miss and calculate the checksum on each Data packets and mention ip adresses & the source & destination.

07

Q. Nos.

* By dividing the network into layers:- By the division of the network into layers, where each layer works on a specific task & improves the efficiency.

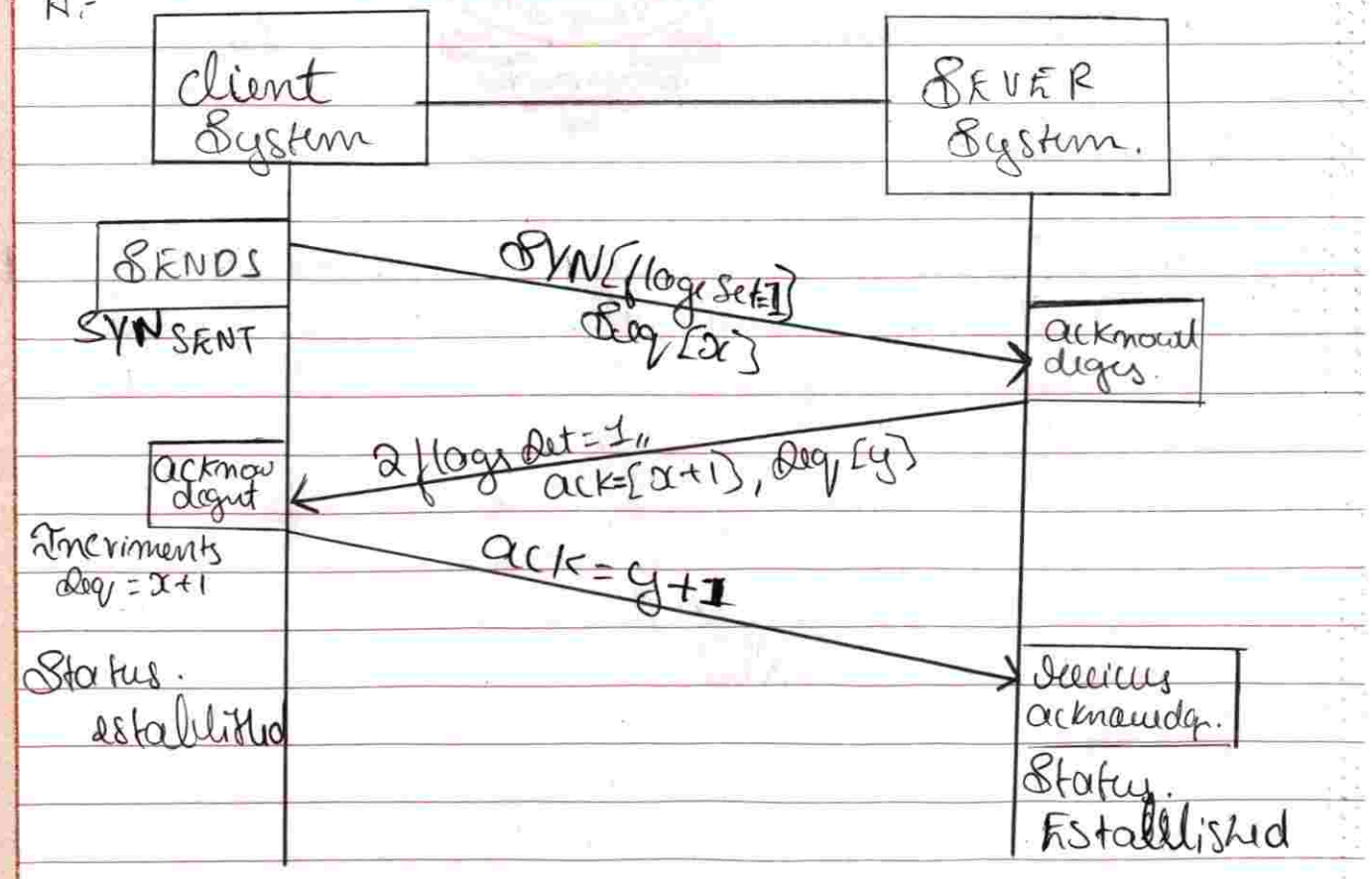
* Bidirectional:- In the transmittion medium it should consist of Bidirectional property where each layer should be ready to perform the one complimentary tasks.

Ex:- the nodes most be ready for either recieving and sending the Data at Both the ends.

Module-4

Q8 a)

A:-



Q. Nos.

TCP Connection Establishment:-

The Connection is being established through a process called as three-way handshake. TCP.

* Where here the Sender from the client side initiate the transmission By generating a Control Segment with SYN[Synchronization flag] Set to 1 & gives randomly a Initial Sequence number to its Data Stream. Set it be Seq = x, Status is said to be SYN SENT.

* On receiving side the SERVER System ~~for~~ receives the this SYN flag and Seq & INS[Initial Sequence number] and generate a buffer memory for it.

* Then On receiving side. the two flages are set to 1 ACK = x+1 & SYN = Y, and sent to the client System. the Status is acknowledgement sent.

* On client side it receives the ~~ack~~ ACK & SYN and increments its Stream to Seq = x+1 and Sends Back a acknowledgement ack = y+1 to the SERVER System.

* After receiving this ACK, the SERVER System Establish the TCP connection and now on Both a Side the connection is being established.

* The Transmission of the Data Can be done.

* After the transmission of the Data, the connection is to be terminated.

* Where again the client sends a SYN to END the the connection. Stays wait.
SEVER

* And the System goes through this SYN and gives an acknowledgement. Back to the client and the System Stays of the System is wait.

* Then again an acknowledgent is being sent By the client to SEVER System. at this point the connection is almost closed. no Data transfer is happening.

* AND the client looks once with the transmission medium whether any data is left and empty it.

* the connection can't be terminated immediately after the acknowledgment it takes time of.
2 MSS 2 MSS.

* This is called as 4 way handShake which is involved in the termination of the connection.

ii) Analyzing the duration if one of the handshake. packets is lost or delayed.

* if any of the packets is lost while transmission sequential transmission is happening.

Q. Nos.

* at a receiver side while transmission the receiver sends an acknowledgment for each Data packets it receives.

* if a packet is being missed also the receiver sends a Duplicate ack to order to re transmit the specific Data packets.

* the lost Data packet is easily recovered due to the initial sequence numbers of the Data packets which helps to arrange them in a perfect sequential order.

Q8

Answer:

A:-

Flow control & Congestion Control mechanism. Used in TCP.

In TCP[Transport layer protol] the problem araise when the Data packets gets collision in between them and lose of packets or the Corruption of the Data packets happens due to this In the transmission medium. So in order to control this collision and fullfill the Draw Backs the Flow control and the congestion control mechanism are being introduced.

Q. Nos.

* Flow control:- the Flow control measure are taken By the intation of the variable ~~is~~ Sliding window where this window make alot of differnce in a transmission, it helps to control the too much of Data overwhelming at the reciver side and causing collisions.

* where it works upon a simple machism, where for each Data packets the reciever side reciver sends the an acknowledgment that How many space is it still available to store the Data packets.

* according to that acknowledgment count the transmission at the sender's sliding window happen where the sliding window dosnot send a data packets above a given range of the acknowledgment from reciver side.

Example: if a Data at the sendes side is about 5000MBS & the reciver sides sends an Acknowledgement after reciving Data that it can store only about 2000MSS, the sends side dosnot send a Data more than the give range of the reciver side to avoid collision

Congestion control:- This is a protocal or a process used to avoid the congestion and the collision among the Data packets while transmision and its working mechanism is ~~the~~ ~~described~~ described as having the windows at Both the sides.

Q. Nos.

* Where initially a variable Cwind (congestion window at the sender side) & rwind (receiving window at the receiving side).

* The formula on which the mechanism works is

The total data that can be sent = min{Cwind, rwind} //

* Here Starting the congestion window is set to one MSS & we have a limit of Slow Start threshold.

* Where for each acknowledgement sent from the receiving side the congestion window is being exponentially grows its transmission rate

Ex: 1, 2, 4, 8, 16 etc...

* At a point when it reaches the Slow start threshold limit the congestion window is being reset where the SSH (Slow start threshold) limit is equals to half of the Cwind.

$$SSH = \frac{Cwind}{2}$$

&

C = reset to Starting.

if the packets are being missed in between the transmission and the congestion just happens mildly. not to heavy the Cwind is again reset when.

$$SSH = \frac{Cwind}{2}$$

&

$$Cwind = SSH_{11}$$

This protocols are being executed to recovery the packets which is lost.

At the transmission layer the Data arrays are handled at the protocols like. Recongonizing the intial sequence number & calculating the checksum and verifying to the check sum before opening the playhood data at the receiving end to check whether the Data is corrupted or not.

Module-3,,

Q6
b
Answer:-

IPV4 Structure:- It is a Structure that consists of the variable to Headers and are less efficient compared to the IPV6, where due to variable Heady size it Harder to look at the exact playhood data. and it consist of the multiple ~~Comments~~ componets through which the transmission can be done.

Q. Nos.

These components are:

Version: 4 bytes

It Specifies the what version of the protocol is being used to conduct the transmission of the IPV4.

Header:
4 bytes:

It consist of the ip source & the destination address and its multiple 5 times to get the original size 5×4=20 bits. and consist of the playhood Data.

protocol:
Size: 8 bits.

This is a Standard Set of instructions used to conduct the transmission.

TTL-Time:
Size: 8 bits:

This is a important component which prevents a Data host in a endless hoop across the network. This Component keep the hop count of the Data packets which is being travelled.

Q. Nos.

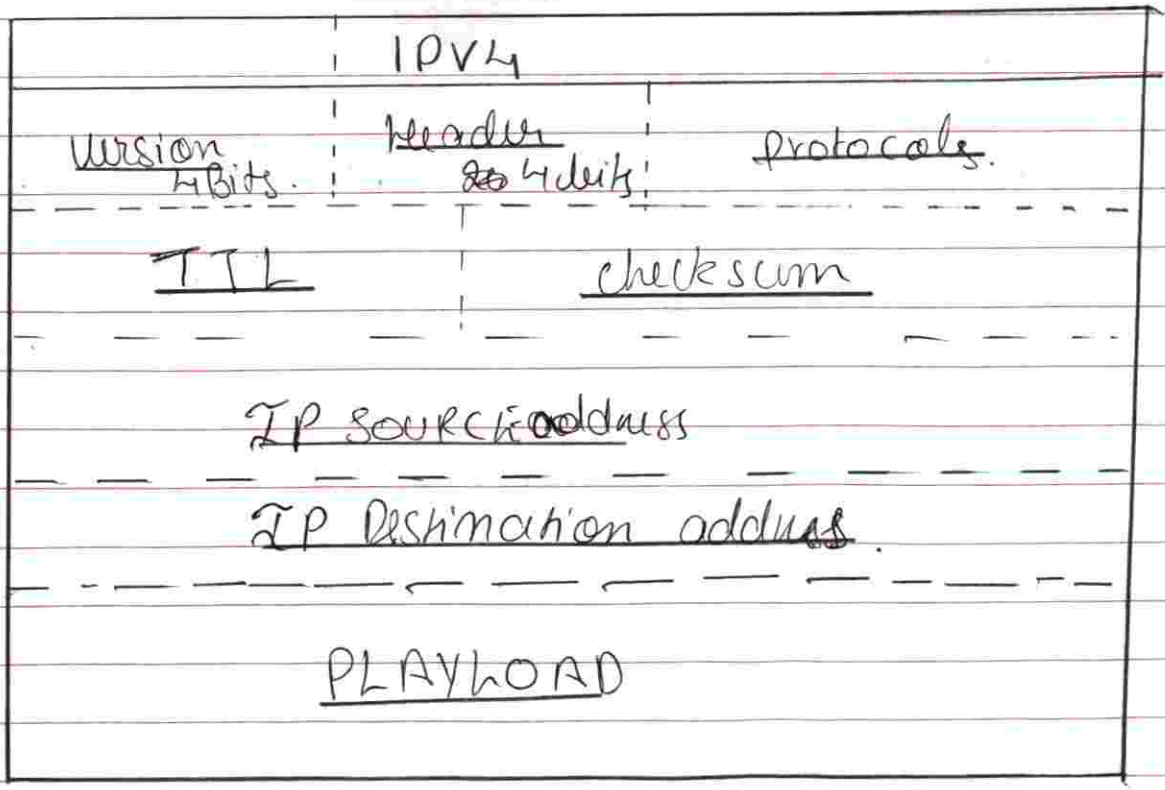
Ip Source address:

* It is the addres of the source from which the Data is being transmitted.

Ip destination address:

* It is the addres of the destination where the Data packets have to be transmitted.

Checksum:- the check sum calculation is being done on the bits to prevent the bits from getting corrupted when the Both Sender & receiving side must perform the checksum to retrieve get a error free data if the calculation on the checksum dosn't match the Data ~~packeted~~ packet is thrown and requested for retransmission.



Q. Nos.

playload:- the Data which is transfer By the upper layer is called as playload.

* The Data packets are Stamped with the IP address of there Source & destination in their headers where. the header is heavy here

* And the Data passes through the network where the TTL - Time to live & the Compont is essential to track the Data packets across the network so it doesn't ~~Dont~~ Stuck in Endless loop.

* the checksum is being calculated to ensure the Data is Corrupted or not while transmi - Hing.

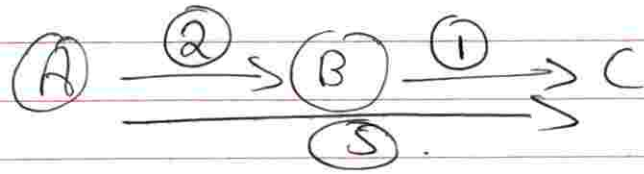
Q6
a Answer:-

Distance vector Routing (DVR):- In DVR the nodes doesn't know the network map of the whole transmission medium what it known is only the nearst neighbour to it and it calculate the distance through that neighbour. and each neighbour Share there own path to the goal node. or a ~~P~~ target node. By having this information & the nodes verify & calculate the shortest path from which the node can reach its targeted destination.

Q. Nos.

Example:-

the node A as to reach the destination node C.



* the path cost from A to C is 5 & the A approach the way from the node B where going through the node B the path cost is equal to 2+1 = 3.

DVR = min(5, 3)
= 3.

* It chooses the path through B because the A as its next neighbour as B & it is the shortest distance.

Algorithm:

Step1: Start
Step2:- Source node & destination node.
Step 3: collect the path cost to the edges & the nods which are neighbours.
Step4: calculate the nearest route
Step5: reach the route by min path cost.
Step6: Stop
Step7: close.

Q. Nos.

module-1

Q1)

ANSWER:

Data Communication: It is the process where the Data packets regardless the types like Binary Bits, making images, or audios are being transmitted across the different computers By the network.

The fundamental characteristics are:-

* Resource Sharing: where it may be the hardware or a Software the resources are being shared like Printer etc....

* portability: This helps in portability where the Data transfer aids in opening or operating the sources in the other version of computers without any issues araised.

* Protocols: the Data communication consist of the number of protocols which are responsible for the error free & collision free data transfer across the network.

Advantages of the Data Communication:-

* Used in the Hospitals: as ECG for the detection of diseases.

Module-2

Q3.

Answer:-

Stop & wait protocols:

The Stop and wait protocols are used to avoid the network jam and the collision occurs the transmission medium where the stop & wait protocols ist check whether the Data packet sent is reached safely to the destination and check the transmission medium is Busy or idel.

* according to this condition the data is being transmitted By this Protocol.

the protocols that comes under this is

PURE ALOHA:- Here the protocols waits until the line to be idel and Starts the trans-mission once the line is free.

Slotted ALOHA:-
Here the time Slotted are introduced data can't be Sent whenever it should wait for a new time Slot. & once the error is caused the whole time slot is wasted.

CSMD:- Here the transmission is made only. when the line is free. but Stile the transimission can happen from Both sides.

Q. Nos.

due to network delay & cause a problem.

CSMPS [with Ensrg].

* it fullfilly the Draw Back of CSMP where the electrical signals are Serized before pasting. the Data Stream.

Q. Nos.

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ROUGH WORK